Harmful Meme Detection in Indic Languages: Dataset Curation and Baseline Development

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Abstract

Memes are increasingly used for communication on social media, but a growing number contain harmful content, including misinformation, stereotypes, and hate speech. In this work, we curate large-scale Indic meme datasets across Hindi, Telugu, Tamil, and Kannada and develop baseline models for zero-shot harmful meme detection. We integrate IndicBERT for textual feature extraction alongside multimodal embeddings and implement a hybrid annotation pipeline combining LLM-assisted reasoning with manual verification. We evaluate our baselines using Macro-F1 to address dataset imbalances and provide insights into multilingual harmful meme detection.

1 Introduction

1.1 Motivation

Memes combine text and visuals and spread rapidly online. While often humorous, many memes are harmful, spreading misinformation or reinforcing stereotypes. Detecting harmful memes is essential for safer digital spaces.

1.2 Relation with NLP

Harmful meme detection requires understanding textual and visual cues, semantic analysis, contextual interpretation, and reasoning. NLP enables extraction and interpretation of textual signals, which are critical for accurate classification.

1.3 Problem Type

This is a binary classification task: predicting whether a meme is harmful or harmless.

2 Feedback and Novelties

2.1 Feedback

Since Presentation 1, our mentor provided guidance on dataset design, model selection, and evaluation strategies. Key feedback includes:

- **Dataset Expansion:** Increase dataset size to ∼8,000 memes per Indic language (Hindi, Telugu, Tamil, Kannada).
- **Annotation Strategy:** Use LLMs for annotation; LLMs struggled with subtle harmful intent, so we tried implementing a hybrid pipeline with manual verification.
- **Textual Feature Extraction:** Replace DistilBERT with IndicBERT for Indian languages. MOMENTA was initially considered, but we refined baselines using IndicBERT and multimodal embeddings.
- **Handling Dataset Imbalances:** Adopt Macro-F1 and category-wise train-test splits to ensure fair representation of harmful and harmless memes.
- **Novelty in model:** Integrate Concept-Net an extension that incorporates back- ground knowledge into the model.

Meeting Summary with Mentor:

Meeting	Discussion Summary
Meeting 1	Discussed the project idea and outlined the overall pipeline.
Meeting 2	Discussed data curation strategies and web-scraping methodology.
Meeting 3	Reviewed project progress and set deadlines for completing dataset curation.
Meeting 4	Explored potential novelties to incorporate and discussed computational requirements.
Meeting 5	Reviewed current progress and fixed the deadline for completing the baseline implementation.

Table 1: Summary of Project Meetings

2.2 Impact and Significance

- **Creation of Linguistic Resources:** Curated large-scale harmful meme datasets for Hindi, Telugu, Tamil, and Kannada. Telugu and Kannada datasets are the first open-source handcrafted datasets for this task.
- **Novel Multimodal Architecture:** Extended MOMENTA by integrating IndicBERT for language-specific embeddings and ConceptNet for external background knowledge, enhancing implicit harm reasoning.
- **Benchmark for Future Research:** Released datasets and baseline results to support reproducibility and provide reference benchmarks for subsequent studies.

2.3 Limitations

- **Performance Gap:** Achieved low accuracy and Macro-F1 scores, often near random prediction, indicating the need for more data and stronger pre-training.
- **Limited Architectural Exploration:** Only one multimodal architecture (MOMENTA-based) was implemented; no comparison with late-fusion models or larger multimodal systems.
- **Lack of Background Context:** ConceptNet frequently failed to retrieve culturally relevant background knowledge, reducing reasoning capability.
- **OCR Fragility Across Scripts:** OCR performance varied across scripts (Hindi, Telugu, Tamil, Kannada), creating inconsistent text extraction quality.
- **Imbalanced Label Distribution:** Class imbalance in some datasets biased the model toward majority labels, lowering Macro-F1.
- **Limited Understanding of Indic Slang:** Non-standard spellings and transliterated slang (Hinglish, Tenglish, etc.) were poorly handled by tokenizers and embeddings.
- **Underutilization of Vision Features:** Stronger vision encoders (e.g., ViT, CLIP large variants) were not explored due to time limits.
- **Unreliable Confidence on Unseen Memes:** For most unseen images, the model outputs harmfulness probabilities in the range of 55–70%, even when the prediction is incorrect. Such moderately high confidence on wrong predictions indicates weak generalization.

2.4 Novelties Introduced

- Curated Indic meme datasets (Telugu, Kannada, Tamil, Hindi).
- Integration of IndicBERT with multimodal embeddings.
- Attempt to gather background knowledge of an image using ConceptNet as an extension to the architecture in MOMENTA paper.

3 Baselines

3.1 State of the Art

Zero-shot harmful meme detection uses large multimodal models (LMMs) with reasoning prompts or retrieval. GPT-4o and Gemini-1.5-Flash are top-performing closed-source models; LLaVA-1.6-34B is a notable open-source baseline used in the paper MIND, which used similarity retrieval and LLM as a judge. The above baselines are only for English text memes but there are no current baselines existing for the indic languages whose datasets, we have made.

3.2 Classic / Training-Based Baselines

Older approaches: multimodal two-stream architectures, late fusion ensembles, and task-specific fine-tuning. These perform well on in-domain data but struggle with out-of-distribution memes.

3.3 Baseline Implementations

No existing implementations for Indic languages. MOMENTA serves as a reference for English.

4 Datasets

- **Dataset Link:** https://iitgnacin-my.sharepoint.com/my?id=%2Fpersonal%2F23110168%5Fiitgn%5Fac%5Fin%2FDocuments%2FMemes%20Dataset&viewid=2a9076ab%2Dacd6%2D4c2f%2Da710%2Dfcd6fe5e872c

4.1 Curated Dataset's Composition and Bias Handling

- **Telugu:** 965 harmful, 1970 harmless (1:2)
- **Kannada:** 573 harmful, 1064 harmless (1:2)
- **Tamil:** 1282 harmful, 1018 harmless (1.26:1)
- **Hindi:** 3776 harmful, 3070 harmless (1.23:1)

Table 2: Dataset composition across train and test splits for each language.

Language	Train (80%)	Test (20%)	Total
Telugu	2348	587	2935
Kannada	1310	327	1637
Tamil	1840	460	2300
Hindi	5477	1369	6846

Bias mitigation: Macro-F1 and category-wise splits to ensure fair evaluation.

4.2 Bottlenecks Encountered

- LLM-based automatic labeling was inaccurate for subtle harmful content.
- Limited publicly available Indic meme datasets.
- Manual curation and annotation were labor-intensive.
- A considerable amount of the memes weren't much clear in terms of resolution because of which text wasn't properly identified, thus, We had to discard them from the training set, which was a waste of time.

4.3 Dataset Extension and Differences

- **Telugu/Kannada:** Handcrafted from scratch, first open-source datasets.
- **Tamil:** Curated from TamilMemes dataset, removed non-harmful troll content.
- **Hindi:** Combined Memotion3 and MIMIC, removed English-only memes, manual annotation, and added web-scraped memes.

5 Experimental Setting

5.1 Overview of Experimental Setting

The goal of the experiment was to classify each meme as **harmful** or **non-harmful**. We curated datasets across four Indic languages and split them into three subsets with the ratio of 80% training, and 20% testing. To minimize potential bias, especially in memes depicting specific individuals, memes referring to the same person were distributed proportionally across the train, and test splits. This ensured fair representation and avoided data leakage through identity overlap.

The experimental pipeline integrates multiple modalities:

- A **CLIP image encoder** for extracting global image-level features.
- An **IndicBERT encoder** for processing and representing the meme text.
- A **VGG-19 model** to extract local region-based features using proposal regions.
- A **ConceptNet-based model** to capture relevant background knowledge and context.

These embeddings were fused using self-attention and cross-attention mechanisms (Code used from MOMENTA paper), followed by a neural network that condensed the multimodal representations into a single binary output corresponding to the harmfulness classification.

5.2 Model Training Setup

We used pre-trained models such as Indic-Bert, CLIP, VGG-19, and ConceptNet as frozen encoders, while the final neural layers were trained on our curated dataset.

The development set was used for hyperparameter tuning, optimizing for both Macro-F1 and accuracy.

Hyperparameters:

The optimizer- Adam
learning rate - 1e-5 (identified as optimal from previous experiments using 15% of train data)
epochs- 40
batch size- 32
Training was performed on Kaggle.

5.3 Reliability and Resource Constraints

Although the results are partially constrained by limited training data and hardware, the experiment maintains interpretability and reliability through its design. Each module has text, background knowledge, and image feature encoders that produce interpretable representations that can be analyzed. Despite resource constraints, the multimodal flow and the interaction between these components provide confidence in the model's decision-making process.

5.4 Models and Approaches Used

Due to limited time, we implemented a single multimodal approach inspired by the **MOMENTA** paper and extended it with ConceptNet embeddings for background knowledge integration. The approach combines textual, visual, and conceptual embeddings using attention-based fusion.

5.5 Approach Variations and Differences

Currently, only one approach has been implemented due to time constraints. Future iterations will involve experimenting with different fusion mechanisms and encoder combinations to compare performance across architectures.

5.6 Architecture and Parameters

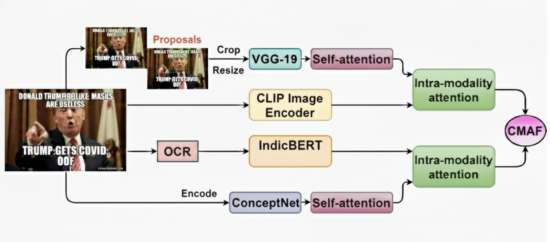


Figure 1: Architecture of the proposed meme classification pipeline.

Parameter Count- **3.74 M** parameters

5.7 Ablation Study

Table 3: Validation Accuracy comparison of both models across Indic datasets.

Dataset	Full Model	Without ConceptNet
Hindi	0.40	0.35
Tamil	0.37	0.31
Telugu	0.36	0.34
Kannada	0.32	0.29

Table 4: Validation Macro-F1 comparison of both models across Indic datasets.

Dataset	Full Model	Without ConceptNet
Hindi	0.41	0.38
Tamil	0.39	0.36
Telugu	0.34	0.30
Kannada	0.33	0.29

5.8 Evaluation Metrics

We used both **Macro-F1** and **Accuracy** as the primary evaluation metrics. Macro-F1 was chosen to handle class imbalance between harmful and harmless memes, while accuracy measures overall correctness. These two metrics together provide a balanced view of the model's classification performance.

5.9 Additional Metrics

No additional metrics were explored, as the combination of Macro-F1 and Accuracy sufficiently captured both class-wise balance and total prediction performance.

5.10 Demonstartion

During the final demonstration, we begin by presenting the raw meme image to illustrate the system's input. The extracted text obtained through OCR is then shown. Next, we display the background knowledge retrieved from ConceptNet, including detected entities and their related concepts. We then present the multimodal fusion stage, where vision features from the MOMENTA backbone, textual embeddings from IndicBERT,

and external knowledge graph embeddings are integrated. Finally, we show the system's prediction output, including harmful and harmless confidence scores and the resulting classification label.

6 Project Management

6.1 Novel Solution Development

We aimed to build upon the existing **MOMENTA** architecture, as it represents a strong multimodal benchmark and provides clear interpretability of the reasoning process behind harmful meme detection. After mentor discussions, we incrementally refined the idea and attempted to integrate Concept-Net an extension that incorporates background knowledge into the model. Instead of using the original MOMENTA pipeline's precomputed background vectors, our approach integrates a knowledge retrieval stage that extracts and embeds background concepts related to entities or objects detected in the meme image. These ConceptNet-derived embeddings were then combined with textual and visual features within the multimodal fusion layer, enhancing contextual understanding and enabling the model to reason about implicit harm or stereotypes.

6.2 Computational Resources

The project required moderate computational support for both training and inference. Experiments were executed on a single workstation with the following configuration:

- **GPU:** 4 x 2 x 15 GB, T4 GPU (Kaggle)
- **CPU:** 4 x 1 cores (Kaggle)
- **RAM:** 4 x 30 GB RAM (Kaggle)
- **Storage:** 80 GB (used)

6.3 Task Distribution

- **Keerthan Varma:** Dataset curation, Documentation, report writing, and Experimentation pipeline.
- **Dinesh Siddhartha:** Dataset curation, ConceptNet integration, and Experimentation pipeline, Web app development.
- **Hemanth:** Dataset curation, Data preprocessing, Report writing and ConceptNet integration.
- **Lakshmi Manasa:** Dataset curation, Evaluation metrics computation, ablation analysis, and report writing.
- **Praveen Kumar:** Dataset curation, Model training, Image Encoder Integration and Conceptnet Integration, Web app development.
- **Bhavana:** Dataset curation, Documentation, report writing, result summarization and Experimentation pipeline.
- **Venkat Akhilesh:** Dataset curation, Hyperparameter tuning, Model training and IndicBert Integration, Web app development.

7 Implementation

What we implemented

- Full pipeline: dataset-prep scripts, OCR/text encoding, train-test 80/20 splits, image encoders, training loop (existing).
- Encoders used: CLIP (global image), VGG-19 (local proposals), IndicBERT (text), ConceptNet embeddings (background knowledge), MOMENTA (fusion and attention modules).

Novelty delivered

- Primary novelty: one of the first curated multilingual Indic meme datasets (Hindi/Telugu/Tamil/Kannada).
- Model novelty: integration of ConceptNet embeddings into a MOMENTA paper's pipeline.

Code and reproducibility

- Repository containing the codebase: `https://github.com/Akhilesh348/Multilingual-Memes-Classification-Harmful-Non-Harmful-.git`.
- Web Application Codebase: `https://github.com/Praveennayak22/Meme_Classifier_Web_App`
- Model Checkpoint: `https://drive.google.com/file/d/1_gywN0IEiFIppZGsIw2Frhbo_XtxZx0q/view?usp=sharing`

Evaluation

- Metrics used: Macro-F1 (to handle class imbalance) and Accuracy.

8 Results and Findings

The model shows consistent performance across languages, capturing multimodal cues despite differences in dataset size and linguistic diversity. However, performance remains below English models (60% accuracy), indicating room for improvement. Our results show that integrating ConceptNet gives the model a clear performance edge, particularly in cases where background knowledge is essential for understanding the meme's context. Overall, while results are promising, they are still far from optimal and random prediction gives approximately 50% accuracy. Stronger pretraining, richer cross-lingual features, and larger data are needed for better outcomes.

8.1 Qualitative Error Analysis

To better understand the behavior of our multimodal classifier, we present three qualitative case studies: a false positive, a false negative, and a correctly classified complex meme. For each example, we analyze why the model behaved the way it did and which components (OCR, IndicBERT, MOMENTA, ConceptNet) contributed to the outcome.

8.1.1 Example 1: False Positive (Harmless Meme Predicted as Harmful)

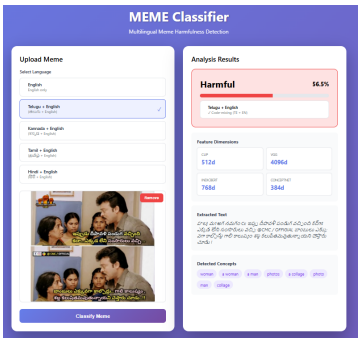


Figure 2: A harmless meme misclassified as harmful.

Ground Truth: Non-Harmful
Model Prediction: Harmful
Confidence: 56.5

Analysis: The model likely failed because the background knowledge of the meme isnt understood and is likely misunderstood by the model. Common causes include:

- OCR extracted noisy text or hallucinated tokens.
- The vision encoder misinterpreted facial expressions or scene tone as aggressive.

8.1.2 Example 2: False Negative (Harmful Meme Predicted as Harmless)

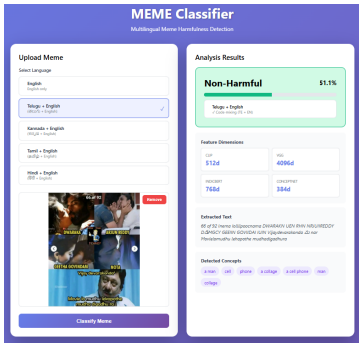


Figure 3: A harmful meme misclassified as harmless.

Ground Truth: Harmful
Model Prediction: Non-Harmful
Confidence: 51.1

Analysis: The misclassification occurred because the model failed to recognize the text's semantic meaning properly and likely didnt identify the actual underlying background knowledge. Typical failure sources:

- Harm expressed through slang is not recognized by IndicBERT.
- ConceptNet lacked connections for the related specific terms like kiss.

8.1.3 Example 3: Correct Prediction on a Complex Meme

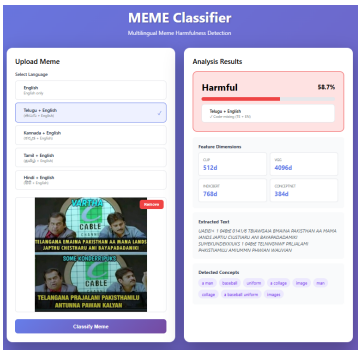


Figure 4: A meme correctly classified by the model.

Ground Truth: Harmful
Model Prediction: Harmful
Confidence: 59

Analysis: The model succeeded because it was able to rightly interpret the entire text in the image. Possible contributing factors:

- IndicBERT captured the nuanced meaning of the meme text.
- ConceptNet added contextual knowledge.
- Fusion of multimodal signals aligned strongly with harmful semantics.

9 Future Work

While the proposed system demonstrates strong performance in multimodal meme classification, several directions can further enhance its robustness and applicability. First, the integration of more sophisticated knowledge graphs such as ATOMIC or multilingual Wikidata can provide deeper contextual grounding, especially for culturally nuanced memes. Second, improving OCR performance for low quality, stylized, or heavily distorted meme text remains an important challenge; incorporating fine-tuning OCR systems on memespecific datasets may significantly reduce information loss. Third, future extensions can explore fine-grained harmfulness categories (like political hate, cyberbullying) instead of binary labels, enabling richer downstream moderation pipelines. Another direction is the adoption of large vision-language models (e.g., LLaVA) for zero-shot and few-shot meme understanding, potentially reducing the reliance on manual feature engineering. Additionally, scaling the dataset across more Indic languages and incorporating code-mixed text can help build a more inclusive and generalizable system. Finally, real-time deployment considerations such as latency optimization, model compression, and user feedback loops can support the development of an efficient and adaptive moderation tool suitable for production environments.

References

- **Momenta:** `https://arxiv.org/pdf/2109.05184`
- **MIND:** `https://arxiv.org/pdf/2507.06908`